Shammas Ahmed

+1 (346) 212-7774 | shammasahmed@rice.edu | linkedin.com/in/shammas-ahmed05 | shammasahmed.github.io

Education

Rice University Houston, TX
B.S. Operations Research (specialization in Algorithms) *May 2027*
GPA: 3.92/4.0

Relevant Coursework
Linear and Integer Programming, Convex Optimization, Discrete Optimization, Large-Scale Optimization
Stochastic Optimization, Online Optimization, Machine Learning with Graphs, Algorithm Design, Simulation Modeling

Experience

DSO Summer Scholar May 2026 – August 2026
Data Science & Operations Department, USC Marshall *Los Angeles, CA*

- Analyzed benchmarking and performance metrics across 24 papers on machine learning for optimization
- Established a constant additive "*look-ahead premium*" (LAP) in regret-based evaluation that distorts comparisons of model performance and benchmark difficulty
- Proposed a regret-reporting policy that accounts for LAP to reflect true model performance on given instances
- Research manuscript accepted at the **NeurIPS MLxOR Workshop 2026**

Undergraduate Research Assistant - Mixed Integer Programming May 2025 – Dec 2025
Operations Research Department, Rice University *Houston, TX*

- Investigated a shortest-path problem with homology constraints (applications in robotics, healthcare, and transportation)
- Reformulated the problem as a network-flow optimization model and established NP-hardness
- Developed a polynomial-time heuristic and characterized instances for which the performance ratio is finite

Teaching Assistant - Introduction to Engineering Computation August 2024 – Present
Rice University School of Engineering and Computing, Rice University *Houston, TX*

- Deliver weekly 1-hour lectures on concepts in math and engineering (e.g. neural networks, first-order optimization methods, root-finding methods) with implementation in MATLAB and Python
- Lead weekly office hours, grade assignments, provide feedback for 20+ students

Projects

Homologous Path Planner | *Python, Gurobi* May 2025 – Dec 2025

- Designed optimization-based path planner that finds the shortest path under homology constraints
- Transformed paths into flows, implemented harmonic projection constraints on 20×20 grids
- Implemented the optimization model in Gurobi and visualized solutions in Python

Owl Help You Plan | *Python, OR-Tools, Gurobi* February 2024 – April 2024

- Developed back-end optimization model for a vacation itinerary planning application
- Formulated as a Vehicle Routing Problem with constraints on destinations, travel time, cost and trip duration
- Implemented using OR-Tools and Gurobi, visualized on a local website using React

Research Presentations

Regret Metrics in Decision-Focused Learning - DSO Summer Research Showcase 2026
The Inverse of the Integer Knapsack Problem - UT Arlington Math Symposium 2024
The Inverse of the Integer Knapsack Problem - Rice Summer Undergraduate Research Symposium 2024

Skills

Programming: Python, MATLAB, R
Optimization: Gurobi, JuMP, PyEPO, PuLP, OR-Tools
Developer Tools: Git, GitHub, VS Code, Google Colab
Methods: Mathematical modeling, algorithm design, data analysis, data visualization